

Study Guide, Exercises, and Sample Questions

CSE 4415: Wireless Networks

Islamic University of Technology (IUT)

Lecture 7.2: MAC Efficiency Boost — Achieving Collision-Free Channel Access Using Off-the-Shelf Technologies

1. Topics Covered

- Recap of Solution Approach 1 (Frequency-Domain Backoff, Lecture 7.1) as the baseline against which the two new approaches in this lecture are compared.
- **Solution Approach 2 - Perfect Coordination (PC) protocol:** a distributed protocol that builds a collision-free TDMA schedule purely through repeated learning cycles, at the cost of a dedicated (and fairly complex) learning phase.
- PC's three-stage learning-phase structure, illustrated for $N = 3$ known stations and $K = 4$ available slots per cycle ($N^{max} = 4$):
 - *Learning-to-Win*: all still-pending (undecided) stations participate every cycle; the set of stations active in each slot is observed by everyone, and exactly one station is confirmed as that cycle's winner.
 - *Rectifying-the-Count*: only the already-declared winners participate, re-transmitting to confirm and order the winning set.
 - *Learning-the-Losers*: winners *and* still-pending stations participate together, so that the stations who lost this cycle can be identified and recycled into the next cycle's Learning-to-Win stage.
- How the three sub-phases repeat, cycle after cycle, until every station's rank in the collision-free order is known to the whole network (illustrated over 5 cycles for $N = 3$).
- The Transmission Phase: once every station knows the full winner order (e.g., 1, 3, 2, 1, 3, 2, ... for $N = 3$), stations transmit back-to-back in that fixed order with zero collisions and zero idle time.
- PC protocol strengths: fully distributed; highly efficient (no collisions, minimal wasted time) once learning finishes; no idle sensing time before transmission (unlike 802.11 backoff); fast convergence; no special PHY (OFDM) or extra antenna required.
- PC protocol weaknesses: requires tight time synchronization across all nodes; needs a dedicated and *complex* learning phase; inefficient in highly dynamic networks, since every node arrival/departure forces the learning phase to be repeated.
- **Solution Approach 3.1 — Deterministic Backoff / CSMA/ECA:** after a station's *first* successful transmission (won via ordinary random backoff), it stops drawing new random numbers and instead reuses one fixed, pre-computed deterministic backoff value, B_d , after every subsequent success.

- The deterministic-backoff formula: $B_d = \left\lceil \frac{2^k \text{CW}_{\min}}{2} \right\rceil - 1$, evaluated with $k = 0$ immediately after a successful transmission (the retry counter k is reset to 0 on success), so in practice $B_d = \lceil \text{CW}_{\min}/2 \rceil - 1$ is a single fixed constant for a given $\text{CW}_{\min}$.
- Worked numerical instance from the slides: $\text{CW}_{\min} = 16 \Rightarrow B_d = \lceil 16/2 \rceil - 1 = 7$.
- Full walk-through with 4 stations (STA1–STA4, initial random backoffs 6, 11, 1, 1): the initial collision between STA3 and STA4 (both counted down to 0 together), their random re-backoff after collision, the sequence of successes, and how, after each station has won *once*, the network self-organizes into a perfectly periodic, collision-free round of transmissions spaced $B_d + 1$ slots apart — with no central coordination and no dedicated learning phase.
- CSMA/ECA extensions mentioned by name (not derived numerically in the slides): *Hysteresis* and *Fair Share*, used to keep the collision-free schedule stable when many contenders are present, and coexistence/backward compatibility with plain CSMA/CA stations.
- Deterministic Backoff strengths: very simple; no OFDM PHY or extra antenna needed; no tight time synchronization needed; no dedicated learning phase; simple schedule preparation; reasonably fast convergence; scalable; trivial handling of node arrival/departure; backward compatible with CSMA/CA; zero collisions and minimal wastage once converged.
- Deterministic Backoff weaknesses: throughput, while much better than pure random access, may not be fully optimal; a highly dynamic network (frequent new arrivals) can repeatedly disrupt the converged steady state.
- **Solution Approach 3.2 — Beacon-Assisted Deterministic Backoff:** an access point periodically broadcasts a beacon; stations use the beacon as a synchronization point that lets them enter (and later re-enter) a collision-free state with far fewer wasted idle slots, independent of the number of stations or their traffic load.
- Worked numerical instance from the slides: a fixed window $V(d) = 16$ slots. In the first window, station 3 collides with station 1 in slot 1; stations 2, 4, 5 win in later random (“rand.”) slots. From the next beacon onward, every station that has already won transmits in a fixed *deterministic* (“det.”) slot corresponding to its learned rank, while stations that have not yet won keep contending randomly (“rand.”) for the remaining slots — so the collision-free set grows beacon by beacon (window 2: stations 1, 2 deterministic, 3, 4, 5 still random/newly deterministic; by window 3 all five stations are deterministic) until, after a beacon re-synchronizes the schedule, the network reaches a stable, fully collision-free periodic order (2, 3, 4, 5, 1 repeating), after which the window can even shrink (e.g., from $V(d)=16$ down to $V(d)=8$) because fewer random slots are needed.
- Beacon-Assisted Deterministic Backoff strengths: same simplicity/no-special-hardware advantages as plain Deterministic Backoff, *plus* less channel wastage and easier re-synchronization thanks to the periodic beacon.

- Beacon-Assisted Deterministic Backoff weaknesses: same dynamic-network sensitivity as plain Deterministic Backoff, *plus* it needs a central coordinator/AP to transmit the beacon (so it is not fully distributed).

2. Focus Areas

- Being able to name, in order, the three PC sub-phases (Learning-to-Win $\rightarrow$ Rectifying-the-Count $\rightarrow$ Learning-the-Losers) and state *who participates* in each one (pending only; winners only; winners + pending).
- Understanding *why* PC needs three separate sub-phases rather than one: a single phase can identify *a* winner, but cannot simultaneously (a) confirm the winner's exact rank and (b) tell the losers who they are, without additional feedback rounds.
- Contrasting PC's per-cycle, multi-round learning cost against Deterministic Backoff's near-zero learning cost (Deterministic Backoff needs no separate learning phase at all — the schedule emerges automatically from ordinary CSMA/CA behaviour).
- The single most important formula in this lecture: $B_d = \lceil CW_{\min}/2 \rceil - 1$, and the fact that it is evaluated *once*, right after a station's first success, and then reused forever (until a new collision forces a temporary return to random backoff).
- Being able to simulate, by hand, several stations running Deterministic Backoff: tracking each station's countdown counter slot by slot, detecting collisions (two counters reaching 0 in the same slot), applying random re-backoff after a collision, and switching a station to its fixed B_d immediately after its first success.
- Recognizing the emergent pattern: once *every* station has experienced one success, all stations use the identical constant B_d , so the network locks into a fixed, collision-free round-robin cycle of length (number of stations) $\times (B_d + 1)$ slots — with no further collisions unless a new station joins.
- Understanding the role of the beacon in Beacon-Assisted Deterministic Backoff: it does not itself resolve contention; it simply re-synchronizes all stations to a common window boundary, letting deterministic (already-won) stations and random (not-yet-won) stations coexist cleanly within the same $V(d)$ -slot window.
- Being able to explain *why* $V(d)$ can shrink once the network reaches steady state (fewer or zero stations still need random slots, so fewer slots are wasted waiting for them).
- Comparing all three protocols in this lecture (PC, Deterministic Backoff, Beacon-Assisted Deterministic Backoff) against each other and against the Lecture 7.1 Frequency-Domain Backoff scheme, along the same five axes used throughout the course: need for special PHY/OFDM, need for extra antennas, need for tight time synchronization, need for a central coordinator, and behaviour under dynamic (churning) membership.
- Being able to reproduce, from memory, the full strengths/weaknesses lists for PC, Deterministic Backoff, and Beacon-Assisted Deterministic Backoff.

3. How to Study These Topics

1. Redraw the PC protocol's Fig. 1 timeline ($N = 3$, $K = 4$) from memory: label every "null", every single-station slot, and every collision slot across all 5 learning cycles, then write out the final Transmission Phase order it produces (1, 3, 2, 1, 3, 2, ...).
2. Redraw PC's Fig. 2 timeline ($N = 3$, $K = 4$, $N^{max} = 4$), this time explicitly separating each cycle into its three labelled sub-phases (Learning-to-Win / Rectifying-the-Count / Learning-the-Losers), and confirm which stations participate in each sub-phase.
3. Re-derive $B_d = \lceil CW_{min}/2 \rceil - 1$ for several values of CW_{min} (e.g. 8, 16, 32, 64) until the formula is automatic.
4. Hand-simulate the 4-station Deterministic Backoff example from the slides (initial backoffs 6, 11, 1, 1, $CW_{min} = 16$) slot by slot on paper; check every collision and every success against the worked example in Section 4 of this guide.
5. Hand-simulate a *new* set of initial backoffs of your own choosing (see Exercise 2.7) to confirm you can apply the algorithm without copying the slide numbers.
6. Redraw the Beacon-Assisted timeline (windows of $V(d) = 16$ slots) and track, window by window, which stations are "rand." and which are "det.", until you reach the fully converged stable state.

4. Exercises

4.1. Perfect Coordination (PC) Protocol

1. In PC's Learning-to-Win sub-phase for $N = 3$, cycle 1 proceeds slot by slot as: null $\rightarrow \{1\} \rightarrow \{1, 2\} \rightarrow \{1, 3\}$. (a) Which station is confirmed as cycle 1's winner, and at which slot? (b) Why do slots $\{1, 2\}$ and $\{1, 3\}$ *not* themselves produce a winner? (c) What must the network do about stations 2 and 3 next?
2. After cycle 1 identifies station 1 as the winner, the Rectifying-the-Count sub-phase consists of a single slot in which only station 1 transmits. (a) What piece of information does this single slot confirm that the Learning-to-Win sub-phase alone could not? (b) Why is it safe for only the winner(s)-so-far to participate here?
3. The Learning-the-Losers sub-phase for cycle 1 has all three stations (1, 2, 3) transmit together in one slot. (a) Given that the network already knows station 1 won, what new information does this slot provide about stations 2 and 3? (b) Why can this information not simply be inferred from cycle 1's Learning-to-Win slots alone?
4. Cycle 2's Learning-to-Win sub-phase (now only stations 2 and 3 are pending) proceeds as: null $\rightarrow \{3\} \rightarrow \{3, 2\}$. (a) Who wins cycle 2? (b) Compare the number of slots needed in cycle 2 (3 slots) against cycle 1 (4 slots, including the initial null) — why does the pending set shrinking from 3 to 2 stations reduce the number of slots needed?
5. By the end of cycle 3, the full winner order for $N = 3$ stations is known to be 1, 3, 2 (station 1 first, then 3, then 2). (a) What sequence of transmissions occurs during the Transmission Phase for a round of 6 back-to-back packets? (b) Why are there

zero idle slots and zero collisions during this phase, in contrast to every one of the preceding learning cycles?

6. Suppose a PC-based network has $N = 5$ known stations and each learning cycle takes, on average, 4 slots across all three sub-phases combined, with slot duration $9\mu\text{s}$. (a) Compute the total learning-phase overhead in microseconds if the network needs $N = 5$ full cycles (one per station) to fully order all stations. (b) If each subsequent data transmission takes $250\mu\text{s}$ and the network then sends 50 back-to-back packets using the learned order, compute the overall efficiency $\eta = \frac{50 \times 250\mu\text{s}}{(\text{learning overhead}) + 50 \times 250\mu\text{s}}$.
7. Conceptually, why does PC's Learning-the-Losers sub-phase require *both* winners *and* pending stations to transmit together, rather than just the pending stations alone?

4.2. Deterministic Backoff (CSMA/ECA)

8. Compute $B_d = \lceil \text{CW}_{\min}/2 \rceil - 1$ for $\text{CW}_{\min} \in \{8, 16, 32, 64, 128\}$. Tabulate your results.
9. Reproduce, on paper, the slide's 4-station example: initial backoffs $B_1 = 6$, $B_2 = 11$, $B_3 = 1$, $B_4 = 1$, with $\text{CW}_{\min} = 16$ (so $B_d = 7$). (a) At which slot do STA3 and STA4 collide? (c) Which station is the very first to achieve a successful transmission, and at which slot count (relative to the start)?
10. Continuing Exercise 9: after STA4's first success, it reloads its backoff counter to $B_d = 7$. The slides show STA1 achieving its first success shortly afterward, also reloading to $B_d = 7$. (a) Explain why, once *all four* stations have each achieved one success, their transmissions settle into the perfectly periodic 4-station round-robin pattern shown in slide 55's diagram (arrows of length $B_d + 1 = 8$ slots). (b) What is the period, in slots, of one complete round through all 4 stations once converged?
11. Using the worked example in Section 4 (stations A, B, C; $\text{CW}_{\min} = 8$; $B_d = 3$): (a) verify by direct counting that A, B, and C each achieve their first success within the first 8 slots. (b) Confirm the slot at which B and C's counters first coincide at 0 (i.e., re-derive the collision at slot 8 shown in the table).
12. For the same A/B/C example, suppose instead that at slot 4 station B's original random draw had been $B_B = 4$ instead of 5 (all else unchanged: A=3, C=0). Re-run the countdown table from slot 0 and determine (a) whether a different pair of stations collides, and if so, at which slot; (b) whether the eventual periodic pattern (once all stations have succeeded once) still has period $B_d + 1 = 4$ slots for each station.
13. A network of 6 stations uses Deterministic Backoff with $\text{CW}_{\min} = 32$. (a) Compute B_d . (b) Once fully converged (every station has won once and no further collisions occur), how many slots does one complete round-robin cycle through all 6 stations take? (c) With a slot duration of $9\mu\text{s}$ and a data+ACK duration of $250\mu\text{s}$ per station per round, compute the channel efficiency of one full converged round, $\eta = \frac{6 \times 250\mu\text{s}}{6 \times 250\mu\text{s} + 6 \times (B_d + 1) \times 9\mu\text{s}}$.
14. **(Design your own instance.)** Pick any three distinct initial random backoffs of your own for three stations X, Y, Z with $\text{CW}_{\min} = 16$ (so $B_d = 7$), such that at least one collision occurs among the very first countdown before every station has

won once. Hand-simulate the full countdown table (in the style of Section 4) until all three stations have succeeded at least once, and state the final steady-state round-robin order.

15. Explain, using the B_d formula, why a *larger* $CW_{\min}$ produces a *longer* steady-state period per station (more idle slots between two successes of the same station), and why a network designer might therefore prefer a smaller $CW_{\min}$ once the network has converged — what would the corresponding risk be if $CW_{\min}$ were set too small to begin with (i.e., before convergence)?

4.3. Beacon-Assisted Deterministic Backoff

16. In the lecture's example, $V(d) = 16$ and, in window 1, station 3 collides with station 1 in slot 1, while stations 2, 4, and 5 each win their own random slot later in the same window. (a) How many stations have *not yet* won by the end of window 1? (b) In window 2, those un-won stations (1 and 3) contend again as "rand.", while stations 2, 4, 5 now transmit in fixed "det." slots — why can the already-won stations safely use fixed slots without re-contending?
17. By window 3 (slots 33–48 in the lecture figure), *all five* stations transmit as "det." (deterministic) — no "rand." slots remain. (a) What does this imply about the number of wasted/idle slots in window 3 compared to window 1? (b) Explain qualitatively why Beacon-Assisted Deterministic Backoff is described as having "less channel wastage" than plain Deterministic Backoff.
18. At slot 49 (window 4), a new beacon arrives. The lecture shows the schedule reorganizing to 2, 3, 4, 5 (deterministic) followed by 1 (also now deterministic, but listed last). (a) Why might the beacon cause a re-ordering rather than simply repeating the exact same sequence as window 3? (b) What role does the beacon play here that a de-centralized protocol (like plain Deterministic Backoff) would not have available?
19. By slots 129–144, the lecture shows the network reaching a "stable state" with order 2, 3, 4, 5, 1 repeating exactly, after which the window size itself shrinks from $V(d) = 16$ down to $V(d) = 8$. (a) Why is it safe to shrink the window once the network is stable? (b) If each slot is $9\mu s$, compute the time saved per window by shrinking from 16 to 8 slots, and the percentage reduction in window duration.
20. A network with $N = 8$ stations uses Beacon-Assisted Deterministic Backoff. Once fully converged (stable state, all 8 stations deterministic), the window shrinks to exactly $V(d) = 8$ slots for signalling, followed by 8 data transmissions of $250\mu s$ each. With slot duration $9\mu s$, compute the channel efficiency of one converged beacon interval, and compare it against the same 8 stations running plain 802.11 DCF with an average temporal backoff of $100\mu s$ per station per packet (no beacon overhead, but full per-packet contention).
21. Compare Beacon-Assisted Deterministic Backoff against the pure PC protocol from Section 5.1 on the specific question of *central coordination*: which of the two protocols is "fully distributed," and which requires an AP? Does requiring an AP make Beacon-Assisted Deterministic Backoff strictly worse, or does it trade one weakness (no coordinator) for a different strength (lower channel wastage, easier re-synchronization)? Justify your answer using specific advantages/weaknesses from Section 1.

4.4. Comparative Analysis

22. Build a comparison table with rows {Frequency-Domain Backoff (Lecture 7.1), PC protocol, Deterministic Backoff, Beacon-Assisted Deterministic Backoff, plain 802.11 DCF} and columns {needs special PHY/OFDM?, needs extra antenna?, needs tight time synchronization?, needs a central coordinator/AP?, handles dynamic membership well?}. Fill in every cell using only facts stated in the two lectures.
23. All three protocols introduced across Lectures 7.1 and 7.2 (Frequency-Domain Backoff, PC, Deterministic Backoff) are described as “fully distributed,” while Beacon-Assisted Deterministic Backoff is not. For each of the three fully distributed protocols, identify the *other* resource or assumption it substitutes in place of a central coordinator (e.g., an extra antenna, a complex learning phase, or reliance on randomness alone).
24. Rank the four collision-free protocols (PC, Deterministic Backoff, Beacon-Assisted Deterministic Backoff, and Frequency-Domain Backoff) from *least* to *most* disruption caused by a single new station joining an already-converged network. Justify each ranking with one sentence per protocol, citing the relevant weakness from Section 1.
25. A network designer must choose one of the four protocols for a WLAN deployment with (a) no special antenna hardware available, (b) a highly dynamic client population (frequent joins/leaves), and (c) an existing AP that already broadcasts periodic beacons for other purposes. Recommend a protocol and justify your choice using at least three specific facts from Section 1.

5. Sample Questions

5.1. Short Questions

1. Name PC’s three learning sub-phases in order.
2. In PC’s Rectifying-the-Count sub-phase, which stations participate?
3. State the deterministic-backoff formula B_d in terms of $CW_{\min}$ and k , and explain why $k = 0$ is used when computing it.
4. Compute B_d for $CW_{\min} = 32$.
5. What event, in Deterministic Backoff, triggers a station to draw a fresh *random* backoff instead of reusing B_d ?
6. What does the periodic beacon achieve in Beacon-Assisted Deterministic Backoff that plain Deterministic Backoff lacks?
7. In the lecture’s Beacon-Assisted example, what happens to $V(d)$ once the network reaches a stable state, and why?
8. Name the one weakness that PC, Deterministic Backoff, and Beacon-Assisted Deterministic Backoff all three share.
9. Which of the three protocols in this lecture requires a central coordinator (AP)?
10. Name the two named extensions of CSMA/ECA mentioned in the lecture (not derived numerically) and their stated purpose.

5.2. Descriptive Questions

11. Describe, phase by phase, how the PC protocol converts N contending stations with no prior coordination into a fully known, collision-free transmission order, using the $N = 3$, $K = 4$ example from the slides. Explain why each of the three sub-phases is necessary and what would go wrong if any one of them were skipped.
12. Explain the Deterministic Backoff algorithm end to end: the initial random phase, the point at which a station switches to B_d , why B_d is computed with k reset to 0, and why the network nonetheless still experiences occasional collisions even after convergence. Use the worked example in Section 4 to illustrate every stage.
13. Explain how Beacon-Assisted Deterministic Backoff blends random and deterministic access within the same window, how the mix of “rand.”/“det.” stations evolves from the first beacon interval to the stable state, and why the window can shrink once stable.
14. Compare and contrast all three protocols covered in this lecture in terms of (a) how much dedicated signalling/learning overhead they require before reaching a collision-free steady state, and (b) how gracefully they handle a station leaving or a new station joining after convergence.

5.3. Analysis and Design Questions

15. Four stations run Deterministic Backoff with $CW_{\min} = 16$ ($B_d = 7$). Their initial random backoffs are $B_1 = 2$, $B_2 = 9$, $B_3 = 4$, $B_4 = 4$. (a) Hand-simulate the countdown slot by slot until every station has achieved at least one success (note: stations 3 and 4 will collide on their very first countdown — handle this exactly as in the Section 4 worked example, using a fresh draw from $\mathcal{U}[0, 31]$ for the colliding pair). (b) State the final steady-state round-robin order and its period in slots.
16. A campus WLAN is deciding between PC and Beacon-Assisted Deterministic Backoff for a lecture hall with $N = 40$ students’ devices, where roughly 5 devices join or leave every minute (as students arrive late or step out). (a) Using the reasoning from Exercise 6–7 in Section 5.1, argue whether PC’s per-membership-change full re-learning cost would be acceptable here. (b) Argue whether Beacon-Assisted Deterministic Backoff’s incremental (per-beacon-interval) re-convergence would cope better, and why.
17. Suppose a network operator wants to guarantee that Deterministic Backoff never experiences a second collision once every station has had its first success, i.e., no two stations ever reload to a colliding value of B_d at the same slot. Given that B_d is a single shared constant for *all* stations under this scheme, explain why this guarantee is fundamentally impossible without changing the protocol (e.g., without giving each station a distinct deterministic value), and propose one concrete modification to the algorithm that would achieve collision-freedom permanently. Estimate, in one sentence, what new coordination cost your proposed modification would introduce.
18. Design a small numerical scenario (choose your own $CW_{\min}$, number of stations, and initial random backoffs) in which *three* stations, rather than two, collide simultaneously on their very first countdown under Deterministic Backoff. Hand-simulate the

collision-resolution step (all three redraw from the expanded window) and continue the simulation until all three have succeeded at least once.